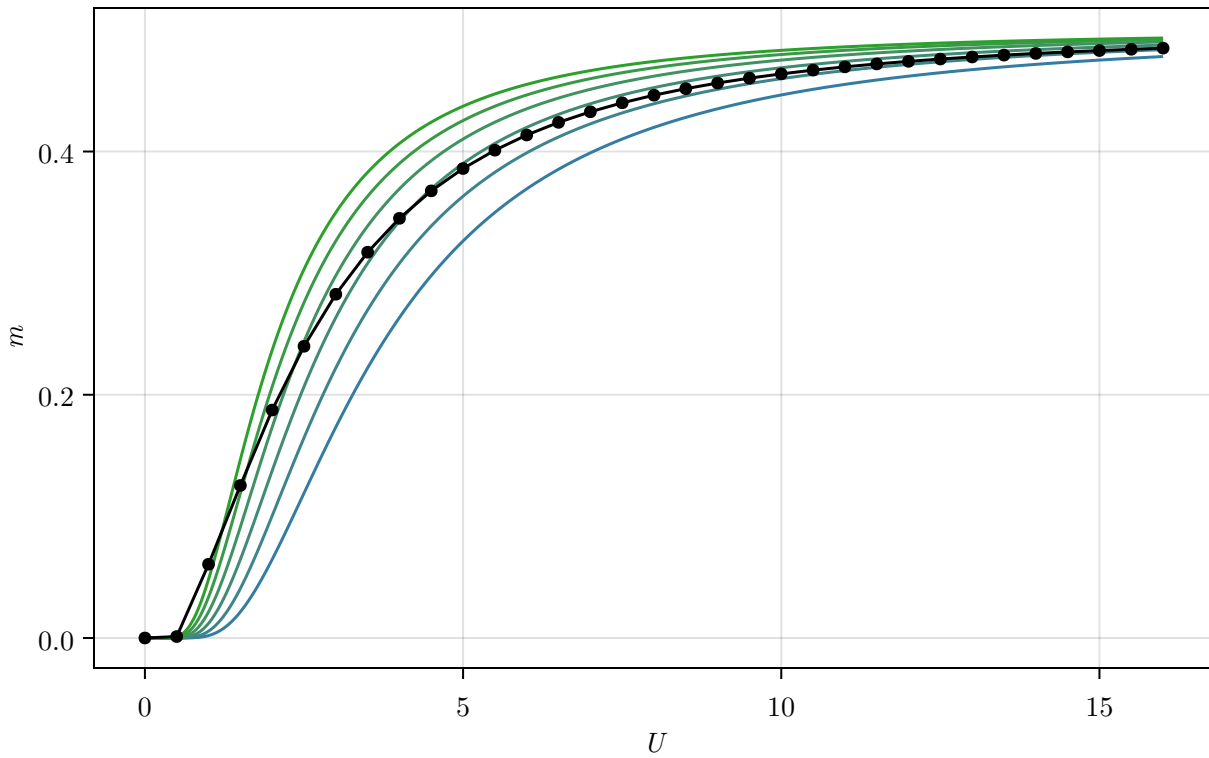


Half-filling hyperbolic approximation



Half-filling exponential approximation

